Cluster 44989

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Subjects

Research fields: Biology, Medicine, Chemistry

The most common research fields of articles in this cluster. Read more.

Research subfields: Virology, Immunology, Microbiology

Common research subfields of articles in this cluster. Read more.

Key concepts: CNS prion disease, prion protein, prion disease pathogenesis, disease prion strains, prion infections

These key concepts are identified algorithmically using article titles and abstracts. Read more.

0.00% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to cybersecurity

Vital signs

24 recent articles in the cluster

24 articles in this cluster were published in the five year period ending in 2023.

Average article is **20.28** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **27.06th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 27.06% of other clusters.

Citation rating: **36.51th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year.In this case, the average article in this cluster is cited more often than roughly 36.51% of all other articles published in the same year.

83.33% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. Read more.

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 19 of 24 articles. 17 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
United Kingdom	4	2.60
United States	4	0.40
Russia	3	0.47
Germany	2	0.50
Nigeria	1	0.50
Taiwan	1	2.67
Spain	1	0.40
Sweden	1	0.25

Top funding countries

Data available for 6 of 24 articles. 6 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
United Kingdom	2	4.95
Japan	1	0.25
Taiwan	1	2.67
Germany	1	0.50

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

- Prion Infectivity and PrPBSE in the Peripheral and Central Nervous System of Cattle 8 Months Post Oral BSE Challenge
  2021: International journal of molecular sciences. 2 citations.
- Potential for transmission of sporadic Creutzfeldt–Jakob disease through peripheral routes
  2021: Laboratory investigation; a journal of technical methods and pathology. 1 citations.
- Transport of Prions in the Peripheral Nervous System: Pathways, Cell Types, and Mechanisms
  2022: Viruses. 4 citations.
- Prion type 2 selection in sporadic Creutzfeldt–Jakob disease affecting peripheral ganglia
  2021: Acta neuropathologica communications. 2 citations.
- Quantitative assessment of the level of biological risk for alimentary-related infections and invasions in the Ryazan region
  2022: Agrarian science. 2 citations.
- The Effects of Immune System Modulation on Prion Disease Susceptibility and Pathogenesis
  2020: International journal of molecular sciences. 17 citations.
- STUDY OF THE MOTIVES OF ORGANIC PRODUCTS CONSUMPTION
  2021: Bulletin of the Moscow University named S U Vitte Series 1 Economics and management. 3 citations.
- Microglia deficiency accelerates prion disease but does not enhance prion accumulation in the brain
  2021: bioRxiv (Cold Spring Harbor Laboratory). 26 citations.
- EFFECT OF FEED ADDITIVE ON MINERAL HOMEOSTASIS OF COWS WITH KETOSIS
  2020: Scientific and Technical Bulletin of State Scientific Research Control Institute of Veterinary Medical Products and Fodder Additives and Institute of Animal Biology. 1 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

- The Effects of Immune System Modulation on Prion Disease Susceptibility and Pathogenesis
  2020: International journal of molecular sciences. 17 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Prion Infectivity and PrPBSE in the Peripheral and Central Nervous System of Cattle 8 Months Post Oral BSE Challenge]
2021: International journal of molecular sciences. 2 citations.

[Transport of Prions in the Peripheral Nervous System: Pathways, Cell Types, and Mechanisms]
2022: Viruses. 4 citations.

[Prion type 2 selection in sporadic Creutzfeldt–Jakob disease affecting peripheral ganglia]
2021: Acta neuropathologica communications. 2 citations.

[THαβ Immunological Pathway as Protective Immune Response against Prion Diseases: An Insight for Prion Infection Therapy]
2022: Viruses. 7 citations.

[Quantitative assessment of the level of biological risk for alimentary-related infections and invasions in the Ryazan region]
2022: Agrarian science. 2 citations.

[The Effects of Immune System Modulation on Prion Disease Susceptibility and Pathogenesis]
2020: International journal of molecular sciences. 17 citations.

[STUDY OF THE MOTIVES OF ORGANIC PRODUCTS CONSUMPTION]
2021: Bulletin of the Moscow University named S U Vitte Series 1 Economics and management. 3 citations.

[Microglia deficiency accelerates prion disease but does not enhance prion accumulation in the brain]
2021: bioRxiv (Cold Spring Harbor Laboratory). 26 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 21 out of 24 articles. 13 of these articles came from these sources.

Source	Articles (last five years)
Viruses	3
International journal of molecular sciences	2
Journal of neuroimmunology	1
PLoS pathogens	1
Laboratory investigation; a journal of technical methods and pathology	1
Neurology	1
bioRxiv (Cold Spring Harbor Laboratory)	1
Bulletin of the Moscow University named S U Vitte Series 1 Economics and management	1
Springer eBooks	1
Access microbiology	1

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 24 of 24 articles. 9 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Neil A. Mabbott	3	11.03
Barry Bradford	3	8.77
I. Petrukh	2	0.20
Adriano Aguzzi	2	0.00
V. V. Vlizlo	2	0.10
D. D. Ostapiv	2	0.10
Barry M. Bradford	2	6.65
Anna Hofmann	2	1.50
Yuliya Kostrova	2	0.38
Petra Schwarz	2	0.00

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 20 of 24 articles. 11 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
University of Edinburgh - United Kingdom	3	3.3
Roslin Institute - United Kingdom	3	3.3
Federal Scientific Agroengineering Center VIM - Russia	2	0.33
University Hospital of Zurich - Switzerland	2	0
Moscow Witte University - Russia	2	0.38
United States Department of Energy - United States	1	0
Fu Jen Catholic University - Taiwan	1	2.67
United States Department of Agriculture - United States	1	0
University of Ibadan - Nigeria	1	0.5
Saarland University - Germany	1	0.5

Organization types

This table covers the 19 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 24 articles total). Read more

Commercial	Education	Nonprofit	Government
0.00%	25.00%	26.25%	10.00%

Top industry organizations

No data available.

Top funders

Data available for 10 out of 24 recent articles. 4 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
Biotechnology and Biological Sciences Research Council - United Kingdom	2	4.95
University of Zurich	1	0.00
Takeda Science Foundation - Japan	1	0.25
WALA Heilmittel GmbH	1	0.50
University Hospital of Zurich	1	0.00
European Research Council - Belgium	1	0.00
Japan Society for the Promotion of Science - Japan	1	0.25
Volkswagen	1	0.50
ACEV Foundation	1	0.00
NOMIS Foundation	1	0.00

Funder types

This table covers the 6 articles in this cluster from the last five years with data about their funding organization types (out of 24 articles total).

Commercial	Education	Nonprofit	Government
0.00%	0.00%	50.00%	50.00%

Patents and industry

51 patents cite articles in this cluster (**68.4th** percentile among all clusters)

4.82% of articles in the cluster are cited by patents

0% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
MOLECULAR ANTIGEN ARRAY	3
PRODUCTS AND METHODS ASSOCIATED WITH MULTIPLE SCLEROSIS AS A TRANSMISSIBLE PROTEIN MISFOLDING DISORDER	3
ANTI-TUMOR/CANCER HETEROLOGOUS ACELLULAR COLLAGENOUS PREPARATIONS AND USES THEREOF	2
METASTASIS INHIBITION PREPARATIONS AND METHODS	2
MYELOID DERIVED SUPPRESSOR CELL INHIBITING AGENTS	2
PREVENTING AND TREATING AMYLOID-BETA DEPOSITION BY STIMULATION OF INNATE IMMUNITY	2
METHODS AND COMPOSITIONS FOR ENHANCING AN IMMUNE RESPONSE, BLOCKING MONOCYTE MIGRATION, AMPLIFYING VACCINE IMMUNITY AND INHIBITING TUMOR GROWTH AND METASTASIS	2
EXTRACELLULAR MATRIX MATERIALS AS VACCINE ADJUVANTS FOR DISEASES ASSOCIATED WITH INFECTIOUS PATHOGENS OR TOXINS	2
MUCOSAL IMMUNIZATION TO PREVENT PRION INFECTION	2

Top patent assignees

Patent assignee	Same-cluster citations
STATE GRID CORPORATION OF CHINA (CHINA)	6
NEW KINPO GROUP	5
WISNIEWSKI THOMAS	4
NOVARTIS (SWITZERLAND)	4
LECHNER FRANZISKA	3
UNIVERSITY OF CALGARY	3
FREY PETER	3
NOVARTIS AG (CH)	3
ORTMANN RAINER	3

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
450	14	prion protein, Prion diseases, PrP, infectious prions, human prion
55390	12	prion diseases, Neurodegenerative Diseases, prion protein, microglia, astrocytes
6746	9	prion protein gene, PRNP gene, Prion Diseases, atypical BSE prions, bovine spongiform encephalopathy
5236	6	Cellular prion protein, prion diseases, prion protein PrPC, PrPC expression, cancer stem cell
23904	6	chronic wasting disease, CWD prions, North American CWD, wasting disease prions, CWD prion detection
5632	3	variant Creutzfeldt-Jakob disease, prion disease, prion protein, Jakob disease cases, prion infectivity
4041	1	prion disease, sporadic Creutzfeldt-Jakob disease, prion protein, sporadic CJD, prion disease surveillance

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
450	18	prion protein, Prion diseases, PrP, infectious prions, human prion
6746	12	prion protein gene, PRNP gene, Prion Diseases, atypical BSE prions, bovine spongiform encephalopathy
5632	12	variant Creutzfeldt-Jakob disease, prion disease, prion protein, Jakob disease cases, prion infectivity
8831	8	murine Creutzfeldt-Jakob disease, prion diseases, solute carrier family, Public Health, disease called kuru
4041	8	prion disease, sporadic Creutzfeldt-Jakob disease, prion protein, sporadic CJD, prion disease surveillance
5236	7	Cellular prion protein, prion diseases, prion protein PrPC, PrPC expression, cancer stem cell
55390	5	prion diseases, Neurodegenerative Diseases, prion protein, microglia, astrocytes
23904	4	chronic wasting disease, CWD prions, North American CWD, wasting disease prions, CWD prion detection
34537	4	prion disease, prion protein gene, PRNP gene, GSS, sporadic Creutzfeldt-Jakob disease
18751	4	mucosal immune system, immune cells, Peyer patches, intestinal epithelial cells, Microfold Cells

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
a2iEditing/RNAEditingIndexer	1	39
gxiaolab/REDITs	1	16
AndraCh/Vacuoles_segmentation	1	0
